Bangladesh Army University of Science and Technology (BAUST), Saidpur

B.Sc. Engineering in Civil Engineering (CE)

List of Courses (Syllabus-2019, Batch 10-11)

Arts & Science Department		Cr.
PHY 1109	Physics I	3.00
PHY 1209	Physics II	3.00
PHY 1210	Physics Sessional	1.50
CHEM 1109	Chemistry	3.00
CHEM 1110	Chemistry Sessional	1.50
MATH 1109	Math I (Differential Calculus, Integral Calculus and Coordinate Geometry)	3.00
MATH 1209	Math II (Ordinary & Partial Differential Equations and Matrices)	3.00
MATH 2109	Math III (Vector Analysis, Laplace transform & Statistics	3.00
HUM 1109	Professional English	2.00
HUM 1110	Technical Report Writing and Presentation	1.50
HUM 1209	Sociology	2.00
HUM 2109	Engineering Economics	2.00
HUM 2209	Financial and Managerial Accounting	2.00
	Arts & Science Total Credit	**30.50**

Inter-disciplinary Department		
Department of CSE		
CSE 2123	Introduction to Computer Programming	3.00
CSE 2124	Introduction to Computer Programming Sessional	1.50
Department of EEE		
EEE 1213	Basic Electrical Engineering	3.00
EEE 1214	Basic Electrical Engineering Sessional	1.50
Department of EEE		
ME 1110	Mechanical Engineering Shop Practice	1.50
	Inter-disciplinary Total Credit	**10.50**

List of Optional Courses (Total Credit-11.00)		
Option 1	***Civil Engineering Practice***	***Cr.***
CE 4203	Socio-economic Aspects of Development Projects	2.00
CE 4205	Business and Career Development	2.00
Option 2	***Structural Engineering***	
CE 4211	Structural Analysis and Design IV	2.00
CE 4213	Prestressed Concrete	2.00
CE 4215	Steel Structures	2.00
CE 4217	Introduction to Finite Element Methods	2.00
CE 4219	Structural Dynamics	2.00
CE 4210	Structural Analysis and Design III Sessional	1.50
CE 4220	Computer Aided Analysis and Design of Structures Sessional	1.50
Option 3	***Geotechnical Engineering***	
CE 4231	Geotechnical Engineering III	2.00
CE 4233	Geotechnical Engineering IV	2.00
CE 4235	Geotechnical Engineering V	2.00
CE 4230	Geotechnical Engineering II Sessional	1.50
Option 4	***Environmental Engineering***	
CE 4241	Environmental Engineering III	2.00
CE 4243	Environmental Pollution Control	2.00
CE 4245	Environmental Development Project	2.00
CE 4240	Environmental Engineering II Sessional	1.50
Option 5	***Transportation Engineering***	
CE 4251	Transportation Engineering III	2.00
CE 4253	Transportation Engineering IV	2.00
CE 4255	Transportation Engineering V	2.00
CE 4250	Transportation Engineering II Sessional	1.50
Option 6	***Water Resources Engineering***	
CE 4261	Water Resources Engineering III	2.00
CE 4263	River Engineering	2.00
CE 4265	Coastal Engineering	2.00
CE 4267	Ground Water Engineering	2.00
CE 4269	Hydraulic Structures	2.00
CE 4260	Water Resources Engineering II Sessional	1.50
	Total Credit Hours	**160.00**

Core Courses of CE Program		Cr.
CE 1100	Civil Engineering Drawing	1.50
CE 1103	Surveying	3.00
CE 1200	Computer Aided Drawing	1.50
CE 1204	Practical Surveying	1.50
CE 1211	Engineering Mechanics	3.00
CE 2100	Details of Construction I	1.50
CE 2105	Engineering Materials	4.00
CE 2106	Engineering Materials Sessional	1.50
CE 2111	Mechanics of Solids I	3.00
CE 2204	Quantity Surveying	1.50
CE 2207	Applied Mathematics for Engineers	3.00
CE 2211	Mechanics of Solids II	3.00
CE 2212	Mechanics of Solids Sessional	1.50
CE 2231	Engineering Geology & Geomorphology	3.00
CE 2261	Mechanics of Fluids	4.00
CE 2262	Mechanics of Fluids Sessional	1.50
CE 3100	Details of Construction II	1.50
CE 3107	Numerical Methods for Engineers	2.00
CE 3108	Engineering Computation Sessional	1.50
CE 4100	Project & Thesis	3.00
CE 4200	Project & Thesis	3.00
CE 4101	Professional Practices and Communication	3.00
CE 4102	Professional Practices and Communication Sessional	1.50
CE 4201	Project Planning and Construction Management	4.00
CE 3111	Structural Analysis and Design I	4.00
CE 3113	Reinforced Concrete Structures I	3.00
CE 3211	Structural Analysis and Design II	3.00
CE 3212	Structural Analysis and Design I Sessional	1.50
CE 3213	Reinforced Concrete Structures II	3.00
CE 3214	Reinforced Concrete Structures Sessional	1.50
CE 4111	Structural Analysis and Design III	3.00
CE 4112	Structural Analysis and Design II Sessional	1.50
CE 3131	Geotechnical Engineering I	3.00
CE 3132	Geotechnical Engineering I Sessional	1.50
CE 3231	Geotechnical Engineering II	3.00
CE 3141	Environmental Engineering I	3.00
CE 3142	Environmental Engineering I Sessional	1.50
CE 4141	Environmental Engineering II	3.00
CE 3251	Transportation Engineering I	3.00
CE 3252	Transportation Engineering I Sessional	1.50
CE 4151	Transportation Engineering II	3.00
CE 3261	Water Resources Engineering I	3.00
CE 3262	Water Resources Engineering I Sessional	1.50
CE 4161	Water Resources Engineering II	3.00
	Core Course Total Credit Hours	**108.00**

Note:

i. Students will take 1 (one) thesis related optional theory course and corresponding 1 (one) sessional course from any elective course groups from option 2 to 6.

ii. Students will also take 3 (three) more optional theory courses from other elective course groups from option 1 to 6, but not more than 1 (one) from any group and 1 (one) more sessional course from any corresponding groups.

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Level 1, Term I		Cr.
CE 1103	Surveying	3.00
PHY 1109	Physics I	3.00
CHEM 1109	Chemistry	3.00
MATH 1109	Math I (Differential Calculus, Integral Calculus and Coordinate Geometry)	3.00
HUM 1109	Professional English	2.00
CE 1100	Civil Engineering Drawing	1.50
CHEM 1110	Chemistry Sessional	1.50
HUM 1110	Technical Report Writing and Presentation	1.50
ME 1110	Mechanical Engineering Shop Practice	1.50
	Semester Total Credits	**20.00**

Level 1, Term- II		Cr.
CE 1211	Engineering Mechanics	3.00
PHY 1209	Physics II	3.00
MATH 1209	Math II (Ordinary & Partial Differential Equations and Matrices)	3.00
HUM 1209	Sociology	2.00
EEE 1213	Basic Electrical Engineering	3.00
CE 1200	Computer Aided Drawing	1.50
CE 1204	Practical Surveying	1.50
PHY 1210	Physics Sessional	1.50
EEE 1214	Basic Electrical Engineering Sessional	1.50
	Semester Total Credits	**20.00**

Level 2, Term I		Cr.
CE 2105	Engineering Materials	4.00
CE 2111	Mechanics of Solids I	3.00
MATH 2109	Math III (Vector Analysis, Laplace Transform and Statistics)	3.00
HUM 2109	Engineering Economics	2.00
CSE 2123	Introduction to Computer Programming	3.00
CE 2100	Details of Construction I	1.50
CE 2106	Engineering Materials Sessional	1.50
CSE 2124	Introduction to Computer Programming Sessional	1.50
	Semester Total Credits	**19.50**

Level 2, Term II		Cr.
CE 2207	Applied Mathematics for Engineers	3.00
CE 2211	Mechanics of Solids II	3.00
CE 2231	Engineering Geology & Geomorphology	3.00
CE 2261	Mechanics of Fluids	4.00
HUM 2209	Financial and Managerial Accounting	2.00
CE 2204	Quantity Surveying	1.50
CE 2212	Mechanics of Solids Sessional	1.50
CE 2262	Mechanics of Fluids Sessional	1.50
	Semester Total Credits	**19.50**

Level 3, Term I		Cr.
CE 3107	Numerical Methods for Engineers	2.00
CE 3111	Structural Analysis and Design I	4.00
CE 3113	Reinforced Concrete Structures I	3.00
CE 3131	Geotechnical Engineering I	3.00
CE 3141	Environmental Engineering I	3.00
CE 3100	Details of Construction II	1.50
CE 3108	Engineering Computation Sessional	1.50
CE 3132	Geotechnical Engineering I Sessional	1.50
CE 3142	Environmental Engineering I Sessional	1.50
	Semester Total Credits	**21.00**

Level 3, Term II		Cr.
CE 3211	Structural Analysis and Design II	3.00
CE 3213	Reinforced Concrete Structures II	3.00
CE 3231	Geotechnical Engineering II	3.00
CE 3251	Transportation Engineering I	3.00
CE 3261	Water Resources Engineering I	3.00
CE 3212	Structural Analysis and Design I Sessional	1.50
CE 3252	Transportation Engineering I Sessional	1.50
CE 3262	Water Resources Engineering I Sessional	1.50
	Semester Total Credits	**21.00**

Level 4, Term I		Cr.
CE 4101	Professional Practices and Communication	3.00
CE 4111	Structural Analysis and Design III	3.00
CE 4141	Environmental Engineering II	3.00
CE 4151	Transportation Engineering II	3.00
CE 4161	Water Resources Engineering II	3.00
CE 4100	Project & Thesis	3.00
CE 4102	Professional Practices and Communication Sessional	1.50
CE 4112	Structural Analysis and Design II Sessional	1.50
	Semester Total Credits	**21.00**

Level 4, Term II		Cr.
CE 4201	Project Planning and Construction Management	4.00
CE 42XX	Elective sessional courses from Option 1 ~ 6	2.00
CE 42XX	Elective sessional courses from Option 1 ~ 6	2.00
CE 42XX	Elective sessional courses from Option 1 ~ 6	2.00
CE 42XX	Elective Theory courses from Option 2 ~ 6 (Thesis related)	2.00
CE 4200	Project & Thesis	3.00
CE 42XX	Elective sessional courses from Option 2 ~ 6 (Thesis related)	1.50
CE 42XX	Elective sessional courses from Option 2 ~ 6	1.50
	Semester Total Credits	**18.00**

ID:

Student Name:

Contact No: